DEEPAK MITTAL

Engineering Lead · Machine Learning · Data Science · Generative AI

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PROFESSIONAL SUMMARY

Engineering Lead with **6+ years** of experience turning business problems into working machine learning solutions. At Persistent Systems, I deliver end-to-end AI/ML engagements for **DataRobot's enterprise clients**, from problem framing and data preparation to model evaluation and client handover. My work spans classification, regression and time-series forecasting in healthcare, government and customer analytics, plus hands-on Generative AI in conversational AI, RAG and natural-language data access. Known for end-to-end delivery ownership and clear communication of model trade-offs.

KEY HIGHLIGHTS

- **7 client AI/ML engagements** delivered since March 2025 across healthcare, government and customer analytics.
- **Production-scale data:** re-engineered a **100M+ row** pipeline to resolve critical out-of-memory failures.
- **Client recognition** in two consecutive quarters, plus an independent endorsement for leadership potential.
- **8 industry certifications**, including Databricks Data Engineer Professional and Machine Learning Professional.

TECHNICAL SKILLS

Languages & Data	Python, SQL, Pandas, Polars, NumPy, Matplotlib, Seaborn
Machine Learning	Classification, Regression, Time-Series Forecasting, Feature Engineering, EDA, Class Imbalance Handling, Threshold Optimization, Model Evaluation & Validation, A/B Testing
Generative AI	RAG, Conversational AI, NLP, Prompt Design, Intent Mapping, Agentic AI, Text-to-Data
Platforms & MLOps	DataRobot, Databricks, Snowflake, AWS, Docker, MLOps
Frameworks & Tools	Scikit-learn, TensorFlow, Jupyter Notebook, PyCharm, Power BI

PROFESSIONAL EXPERIENCE

Persistent Systems | Engineering Lead

Mar 2025 – Present

DataRobot account: delivering AI/ML solutions for DataRobot's enterprise clients, from problem definition to handover.

Mediclinic – Preapproval Insurance Prediction *Healthcare · Classification & Regression*

- **Goal:** predict whether an insurance preapproval request will be approved, and how long approval will take, so hospital teams can anticipate rejections and delays.
- Built a **classification model** for approval outcome and a **regression model** for turnaround time; resolved data-quality issues and addressed class imbalance in the approval data.
- Evaluated both models against the client's business requirements before recommending them for use.

Mediclinic – Patient No-Show Prediction *Healthcare · Classification*

- **Goal:** identify patients likely to miss scheduled appointments, so the hospital can plan schedules and outreach proactively.
- Built **challenger models** and benchmarked them against the existing production model to test whether a replacement would perform better.
- Used **threshold-based evaluation** to show the business the trade-off between catching more no-shows and raising false alarms, and prepared client-facing deployment documentation.

Mediclinic – Patient Volume Forecasting *Healthcare · Time-Series Forecasting*

- **Goal:** forecast patient volumes across hospitals, locations and patient categories to support demand and resource planning.
- Prepared and structured historical patient-volume data at several levels of granularity, then developed and compared forecasting and regression approaches for each segment.

Mediclinic – Maternity Delivery Completion *Healthcare · Classification*

- **Goal:** predict whether an expectant mother will complete her delivery at the hospital, helping identify patients at risk of going elsewhere.
- Built the notebook pipeline from **data validation** through **transformation** to a model-ready training dataset with **13 engineered features**, keeping all source columns available for business lookup while controlling the model feature list.
- Trained and evaluated classification models on DataRobot and assessed their suitability against business requirements.

Large-Scale Data Pipeline Optimization *Data Engineering · Pandas to Polars*

- **Problem:** a production data-preparation notebook was hitting out-of-memory failures as its data grew. Migrated it from pandas to **Polars lazy evaluation**, cell by cell, preserving all business logic.
- Resolved date-format inference errors across **100M+ rows** with explicit parsing, fixed a silent correctness bug in "latest user per reference" logic with deterministic aggregation, and recommended a one-time CSV-to-Parquet conversion.

Ministry of Interior – Hajj Pilgrim AI Chatbot *Government · Conversational AI*

- **Goal:** a conversational assistant giving Hajj pilgrims accurate, easy-to-follow guidance.
- Structured the domain knowledge base, identified user intents and designed response flows, with a focus on clear, context-aware and culturally sensitive answers.

Via Directa – Customer Lifetime Value *Customer Analytics · Regression*

- **Goal:** predict the future value of each customer to help the business prioritize retention and marketing effort.
- Owned the full regression pipeline: data understanding, preprocessing, feature engineering, model selection, validation and documentation.

ED&F Man – Talk To My Data *Generative AI · Natural-Language Analytics*

- **Goal:** let business users ask questions of their data in plain English instead of writing queries.
- Supported development of the working solution and enabled the client team to adopt and use it.

KPIT | Senior Software Engineer

Sep 2023 – Sep 2024

- Independently built and delivered a Python-based application within **three sprints**, improving delivery efficiency by **25%**.
- Designed scalable solutions across two projects and coordinated cross-functional teams on Python integrations.
- Ran an independent feasibility assessment for integrating a Generative AI tool and presented the findings and recommendation to technical and non-technical stakeholders.

LTIMindtree | Senior Consultant

Jul 2020 – Aug 2023

Progressed from Trainee to Consultant to Senior Consultant

- Developed and evaluated predictive models using regression and classification techniques to support business forecasting and decision-making.
- Designed and ran **A/B tests** to measure the impact of features and strategies, and presented findings to stakeholders.
- Performed data mining, cleaning and exploratory analysis across multiple datasets, and contributed to business case studies and model assessments.

PERSONAL PROJECTS

Personal Bank Loan Modeling *Classification · AdaBoost*

- Built a supervised-learning workflow with preprocessing and feature engineering to predict personal-loan outcomes; achieved **precision 0.975, recall 0.964 and F1-score 0.970** with a tuned AdaBoost model.

House Price Prediction *Regression · AWS EC2*

- Applied data cleaning, normalization and feature engineering to build house-price regression models, and **deployed the predictions on AWS EC2**.

AI-Powered Marketing Team (Capstone) *Agentic AI*

- Conceptualized a team of AI agents for marketing analysis, targeting and strategy, with workflows showing how agents can augment human decision-making.

CERTIFICATIONS

Databricks	Data Engineer Professional · ML Professional · Data Engineer Associate · Gen AI Engineer Associate
Microsoft	Fabric Analytics Engineer Associate (DP-600) · Azure AI Fundamentals (AI-900)
Snowflake	SnowPro Specialty: Gen AI (GES-C02)
Neo4j	Neo4j Certified Professional (Fundamentals, Cypher Fundamentals, Data Modeling Fundamentals, Cypher Intermediate Queries)

EDUCATION

B.Tech, Computer Science Engineering

Aug 2016 – May 2020

University of Petroleum and Energy Studies (UPES), Dehradun | CGPA 8.23 / 10

Internship: DCM Data Systems, Trainee (Jun – Jul 2019)